

ROLAND U-110 · HARDWARE VS EMULATION

# Note 43, Partial by Partial

One G2 at full velocity on A. Piano 1, recorded from the machine and rendered from the emulator, aligned to the sample and measured against each other.

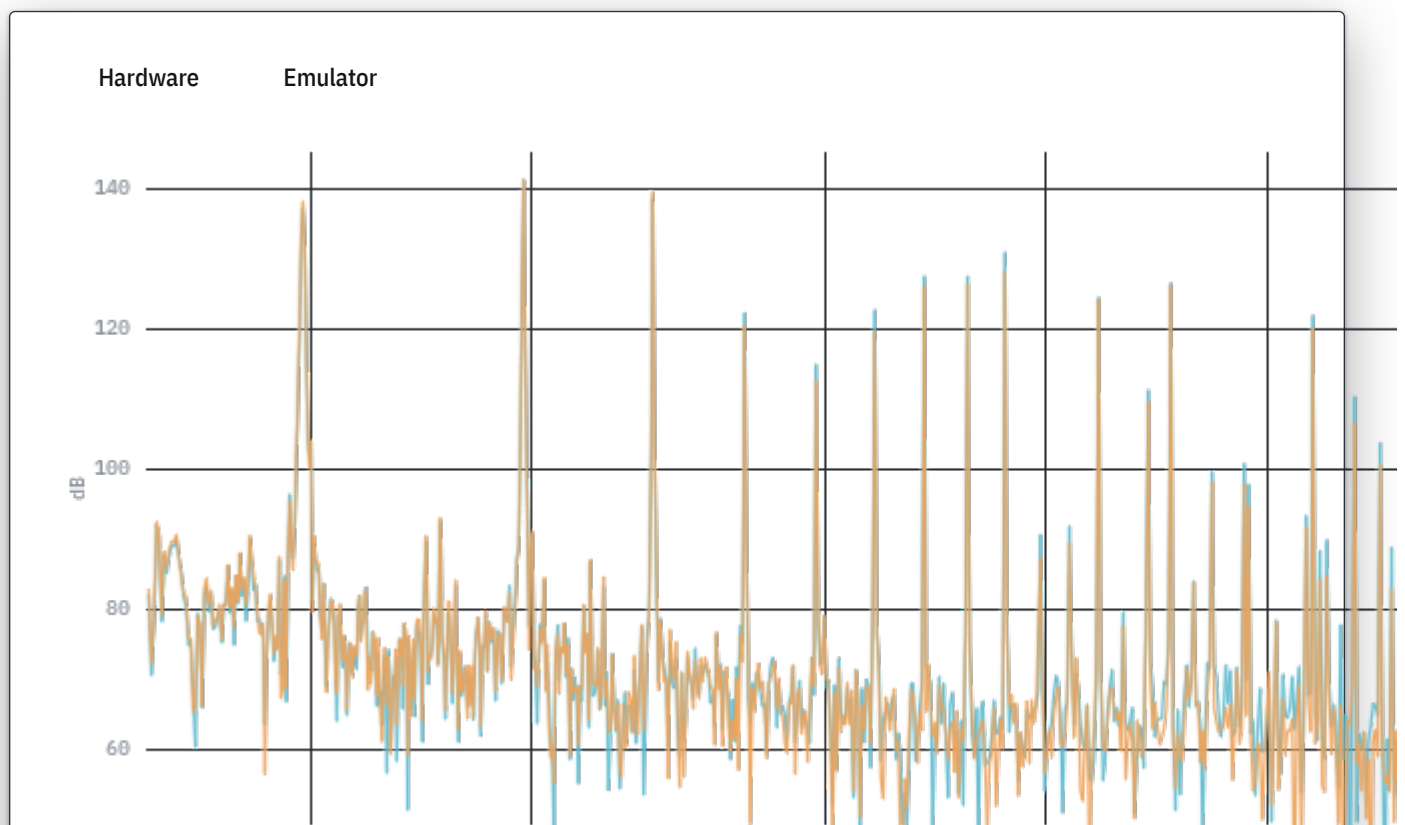
MIDI note 43 · G2 ·  $f_0$  98.00 Hz    Velocity 127    Attack correlation 0.9923  
Alignment 2 samples · 0.04 ms

## What matches

Almost everything. The two waveforms correlate at **0.954** over six seconds, which for independently clocked sources means the emulator is playing the same sample data through the same pitch. Every partial lands within a few cents of its hardware counterpart, and — this is the part worth pausing on — the *inharmonic*ity comes with it. Real piano strings are stiff, so their partials run progressively sharp of true harmonics. The 11th sits +7 cents high, the 18th +25. Both sides do this identically, because it is baked into the sampled waveform.

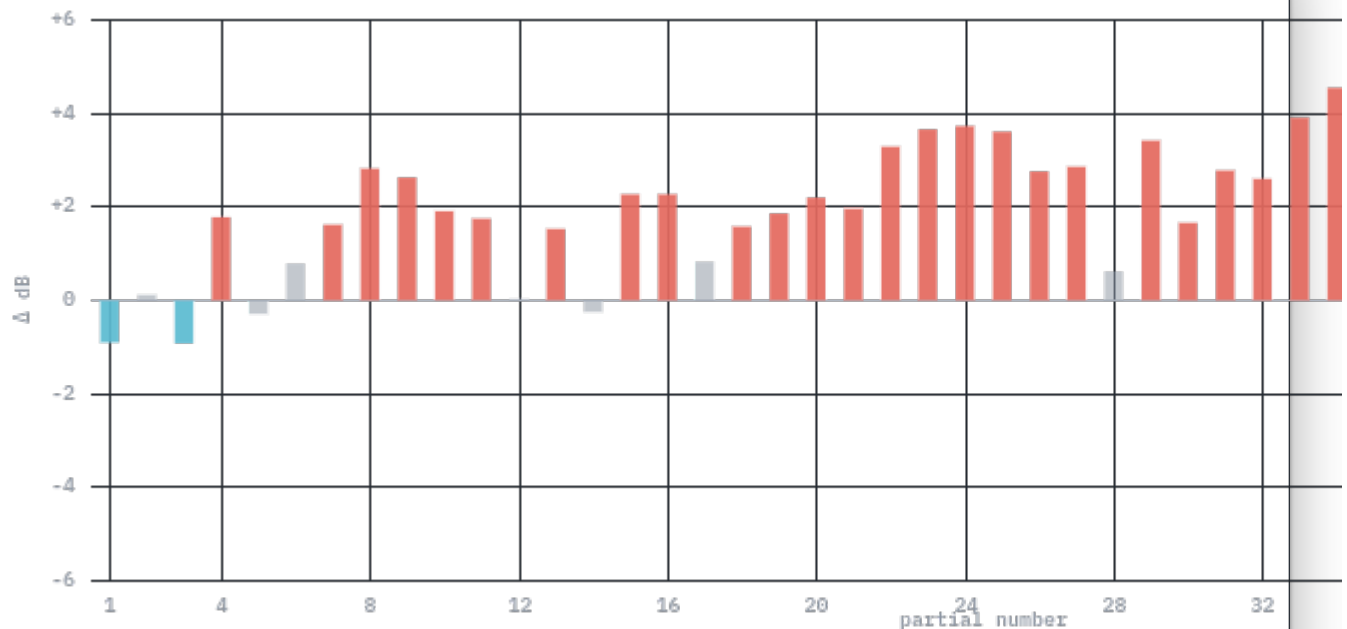
So the difference you are hearing is not pitch, not tuning, and not the sample. It is what happens to those partials over time.

## Spectrum, first two seconds



2.0 s window from 0.15 s after onset, Hann, level-matched on total energy. The traces sit on top of each other through the low partials and separate above roughly 700 Hz.

## Level of each partial

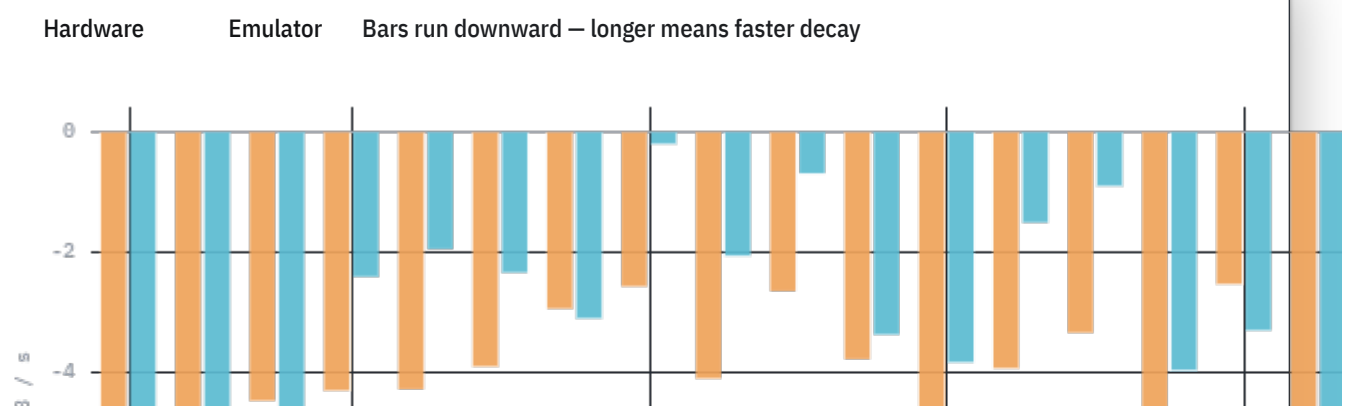


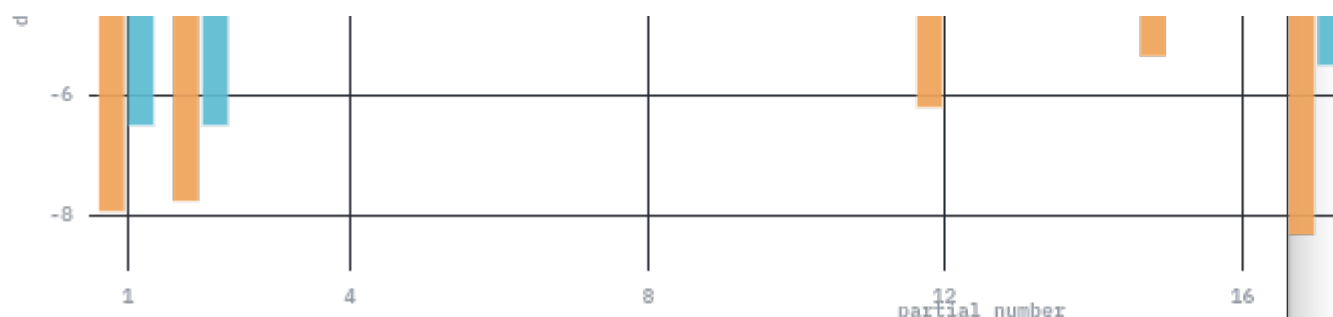
Emulator minus hardware, per partial, over the same window. Below the 4th partial the two agree within a decibel. From the 8th up the emulator runs consistently hot, averaging +2.5 dB across all 48 measured partials.

### The actual difference is decay, not balance

A static +2.5 dB tilt would be an EQ problem, and it would sound like brightness. What the measurement shows instead is that the emulator's partials **fade more slowly than the hardware's** — by 1.3 to 2.6 dB per second on most of them. That is a timbre that changes as the note falls, which is a different thing to hear and a much harder one to ignore.

## Decay rate, partial by partial





Slope of each partial's level between 0.3 s and 4.0 s, in dB per second. The 8th partial is the extreme case: hardware sheds 2.58 dB/s, the emulator 0.21 dB/s – it very nearly holds station for four seconds.

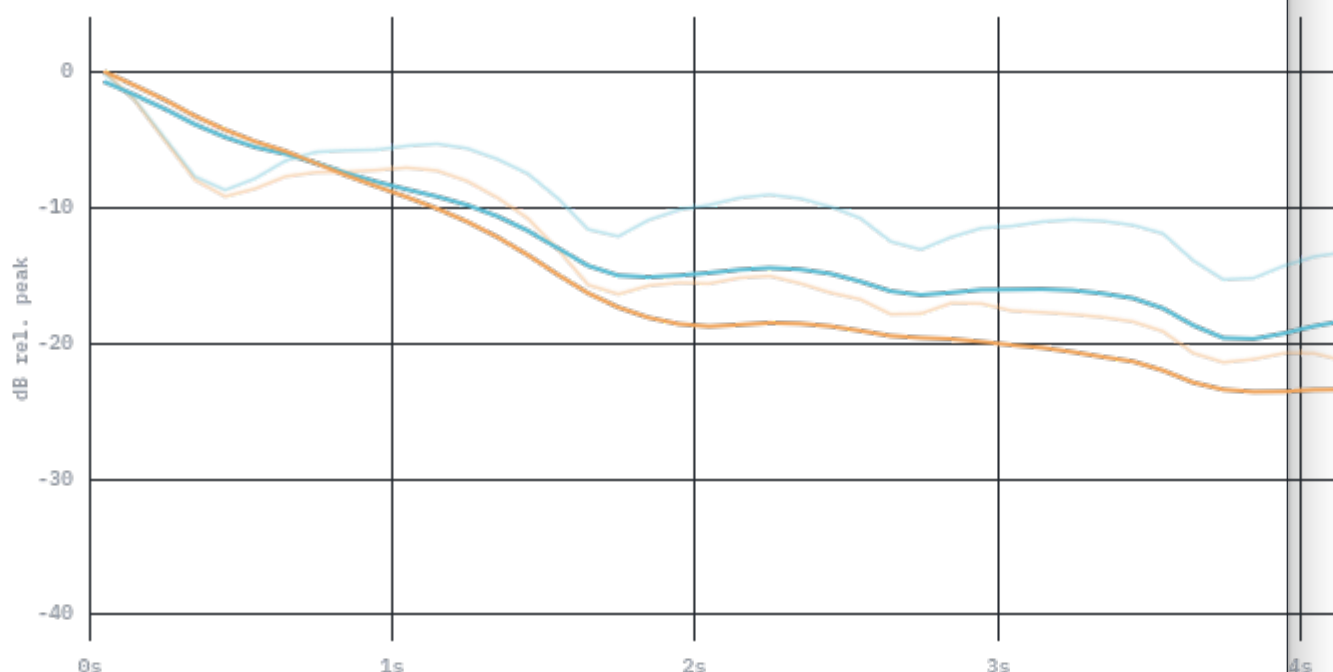
## How the note falls

Hardware, broadband

Emulator, broadband

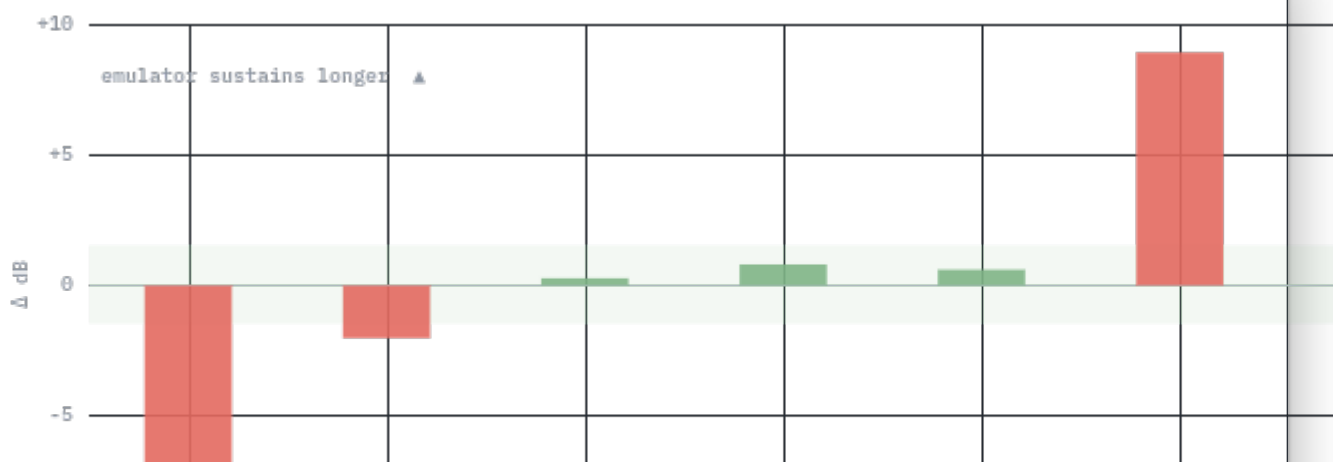
Hardware, 700–2500 Hz

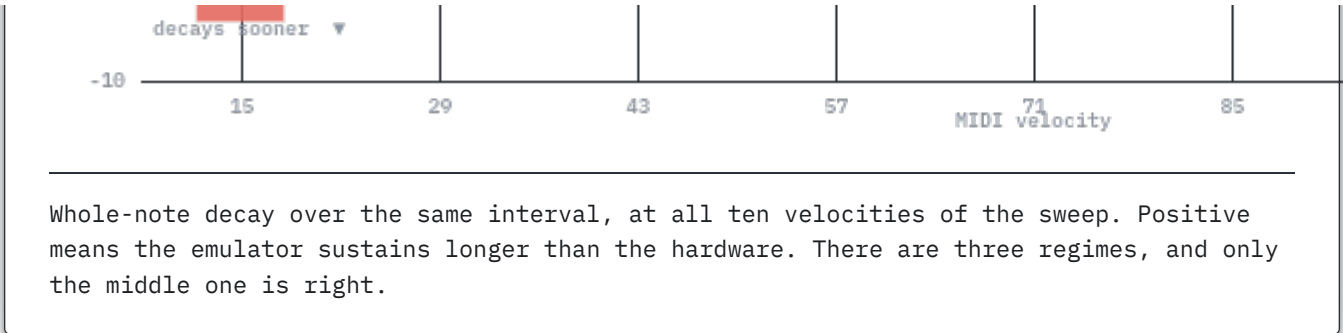
Emulator, 700–2500 Hz



The pale pair is the band where the ear places piano tone. Hardware sheds 11.7 dB there between 0.5 s and 3.75 s; the emulator sheds 5.3 dB. The gap opens from about half a second in and never closes.

## It depends on how hard you play





| VELOCITY 1                | VELOCITY 15-29     | VELOCITY 43-71 | VELOCITY 85-127    |
|---------------------------|--------------------|----------------|--------------------|
| <b>dies early</b>         | <b>2-8 dB fast</b> | <b>±0.8 dB</b> | <b>4-9 dB slow</b> |
| emulator silent by 3.75 s | decays too quickly | correct        | sustains too long  |

## Where this points

The U-110 applies one amplitude envelope per voice, and this note uses two voices with different samples and different envelopes. A per-partial decay error, on a machine with no filter, can only come from those two layers falling at the wrong rates relative to each other — the brighter layer outstaying its welcome shifts the timbre as the note dies.

The velocity dependence is the strongest clue. An error that is absent at velocity 57 and worth 9 dB at velocity 99 is not a filter and not a sample; it is the envelope generator being handed different segment rates at different velocities and getting some of them wrong. That is the ramp generator and the interrupt-driven segment handler, not the output stage.

Not the output EQ. Measured across the same capture, the shipped correction holds the response to within ±0.8 dB from 1 kHz to 8 kHz. This is a separate fault, and it is the one that is audible.

## Partial measurements

| PARTIAL | NOMINAL HZ | MEASURED HZ | INHARMONICITY | HARDWARE DB | EMULATOR DB | Δ     |
|---------|------------|-------------|---------------|-------------|-------------|-------|
| 1       | 98.0       | 98.00       | 0.0 c         | 137.96      | 137.04      | -0.92 |
| 2       | 196.0      | 195.50      | -4.4 c        | 140.96      | 141.06      | +0.10 |
| 3       | 294.0      | 293.00      | -5.9 c        | 139.32      | 138.38      | -0.94 |
| 4       | 392.0      | 390.50      | -6.6 c        | 120.31      | 122.07      | +1.76 |
| 5       | 490.0      | 489.50      | -1.7 c        | 122.42      | 122.11      | -0.31 |
| 6       | 588.0      | 587.50      | -1.5 c        | 126.61      | 127.38      | +0.77 |

|    |        |         |         |        |        |       |
|----|--------|---------|---------|--------|--------|-------|
| 7  | 686.0  | 685.50  | -1.2 c  | 127.46 | 129.06 | +1.60 |
| 8  | 784.0  | 784.50  | +1.1 c  | 126.17 | 128.97 | +2.80 |
| 9  | 882.0  | 882.50  | +1.0 c  | 133.06 | 135.67 | +2.61 |
| 10 | 980.0  | 982.00  | +3.5 c  | 134.58 | 136.47 | +1.89 |
| 11 | 1078.0 | 1082.50 | +7.2 c  | 121.43 | 123.16 | +1.73 |
| 12 | 1176.0 | 1181.00 | +7.4 c  | 127.79 | 127.80 | +0.01 |
| 13 | 1274.0 | 1282.00 | +10.9 c | 121.62 | 123.13 | +1.51 |
| 14 | 1372.0 | 1383.00 | +13.8 c | 121.02 | 120.75 | -0.27 |
| 15 | 1470.0 | 1482.50 | +14.7 c | 126.02 | 128.27 | +2.25 |
| 16 | 1568.0 | 1584.50 | +18.1 c | 118.25 | 120.50 | +2.25 |
| 17 | 1666.0 | 1686.50 | +21.2 c | 116.07 | 116.88 | +0.81 |
| 18 | 1764.0 | 1789.50 | +24.9 c | 123.22 | 124.78 | +1.56 |
| 19 | 1862.0 | 1865.00 | +2.8 c  | 101.30 | 103.14 | +1.84 |
| 20 | 1960.0 | 1964.00 | +3.5 c  | 80.16  | 82.34  | +2.18 |
| 21 | 2058.0 | 2063.50 | +4.6 c  | 100.48 | 102.42 | +1.94 |
| 22 | 2156.0 | 2163.00 | +5.6 c  | 94.12  | 97.39  | +3.27 |
| 23 | 2254.0 | 2263.00 | +6.9 c  | 91.47  | 95.11  | +3.64 |
| 24 | 2352.0 | 2362.50 | +7.7 c  | 85.41  | 89.12  | +3.71 |
| 25 | 2450.0 | 2419.00 | -22.0 c | 110.05 | 113.64 | +3.59 |
| 26 | 2548.0 | 2523.00 | -17.0 c | 98.96  | 101.70 | +2.74 |
| 27 | 2646.0 | 2631.00 | -9.8 c  | 112.13 | 114.97 | +2.84 |
| 28 | 2744.0 | 2741.50 | -1.6 c  | 109.00 | 109.60 | +0.60 |
| 29 | 2842.0 | 2851.50 | +5.8 c  | 106.76 | 110.16 | +3.40 |
| 30 | 2940.0 | 2960.50 | +12.0 c | 94.85  | 96.49  | +1.64 |
| 31 | 3038.0 | 3073.00 | +19.9 c | 110.03 | 112.79 | +2.76 |
| 32 | 3136.0 | 3170.50 | +19.0 c | 103.89 | 106.47 | +2.58 |
| 33 | 3234.0 | 3185.50 | -26.1 c | 106.98 | 110.87 | +3.89 |
| 34 | 3332.0 | 3298.50 | -17.5 c | 112.64 | 117.17 | +4.53 |
| 35 | 3430.0 | 3414.00 | -8.1 c  | 86.17  | 89.69  | +3.52 |
| 36 | 3528.0 | 3530.00 | +1.0 c  | 100.08 | 103.65 | +3.57 |
| 37 | 3626.0 | 3641.00 | +7.2 c  | 100.01 | 104.22 | +4.21 |
| 38 | 3724.0 | 3763.00 | +18.1 c | 96.36  | 101.06 | +4.70 |
| 39 | 3822.0 | 3876.00 | +24.3 c | 104.00 | 107.12 | +3.12 |
| 40 | 3920.0 | 3876.00 | -19.5 c | 104.00 | 107.12 | +3.12 |
| 41 | 4018.0 | 3994.50 | -10.1 c | 100.60 | 104.11 | +3.51 |
| 42 | 4116.0 | 4123.50 | +3.2 c  | 91.33  | 94.53  | +3.20 |

|    |        |         |         |       |       |       |
|----|--------|---------|---------|-------|-------|-------|
| 43 | 4214.0 | 4240.00 | +10.7 c | 88.79 | 92.78 | +3.99 |
| 44 | 4311.9 | 4364.50 | +21.0 c | 82.85 | 87.00 | +4.15 |
| 45 | 4409.9 | 4364.50 | -17.9 c | 82.85 | 87.00 | +4.15 |
| 46 | 4507.9 | 4485.50 | -8.6 c  | 86.95 | 90.71 | +3.76 |
| 47 | 4605.9 | 4612.00 | +2.3 c  | 90.97 | 95.65 | +4.68 |
| 48 | 4703.9 | 4739.00 | +12.9 c | 86.13 | 91.78 | +5.65 |

Source: [listen/4/01\\_piano\\_vel\\_43.wav](#) (hardware, UA-25, 48 kHz) against [listen/4-emu/01\\_piano\\_vel\\_43.wav](#) (MAME, output EQ correction on). Aligned by cross-correlation over the first 300 ms, level-matched on total energy. Decay slopes fitted 0.3–4.0 s. Inharmonicity in cents relative to k·f<sub>0</sub>.